On the Necessary Substructures of Finite Contact Graphs: A Conditional-Existence Account of Why a Finite World Is Structured

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Abstract

We ask a structural question—*why is a world structured at all?*—and answer it with a single hypothesis about graphs. A *contact graph* is a finite weighted graph whose vertices are the parts that have been individuated within a whole and whose edges carry the cost of separating one part from another. We take as our only premise that such a graph is *finite* and that its edge weights are bounded below by a positive constant; we call this the Finiteness Premise. From it alone we derive, as a chain of existence theorems, a ladder of substructures that any finite contact graph must contain. We prove: **(T0)** a strictly positive *floor* on every separating boundary—no two parts are adjacent at zero cost; **(T1)** that each vertex is fixed only by its complement-neighbourhood, so that individuation must proceed by negation and admits no selector without regress; **(T2)** that the *cut-residual*—the inter-part boundary content—is bounded below and is invariant under every relabelling (isomorphism) of the graph, while no single vertex realises it, so the residual is a region and never a point; **(T3)** that this residual is the conserved quantity under all re-encodings of the graph's vertices, the unique invariant from which the relabellable surface derives; **(T4)** that every connected subgraph carries its own such invariant under its automorphisms; **(T5)** that selecting a walk on the graph requires an edge-gating function, without which the next step is undetermined; **(T6)** that the committed-edge count along any walk is monotone, so revisiting a vertex is never returning to a state and resemblance never entails identity; **(T7)** that a coherent family of subgraphs acting as one admits a single gating function on their quotient graph, and only one; and **(T8)** that, because the subgraph invariant is complete, contact graphs are specifiable and hence instantiable—authored finite structures exist—subject to three bounds we prove: their walks are not foreseeable from their specification, an authored copy is a distinct structure, and authorship is limited to the resolution of the floor. We close with a Master Theorem: every substructure T0–T8 is forced for finite graphs and is *not* forced for infinite or zero-floor graphs. Structure is therefore neither imposed on a world nor intrinsic to it independent of its parts; it is the necessary form a world must take to be resolved by finitely many bounded parts. *Structure is the shadow of finitude.* Every claim is proved from the single premise; uncertain material has been omitted, and the interpretations a reader may attach are confined to clearly marked remarks on which no theorem depends.

Keywords: contact graph; finite weighted graph; edge-weight floor; individuation by negation; cut invariant; graph isomorphism invariant; gated walk; monotone walk; quotient graph; conditional existence; necessary substructure; authored structure.

Contents

1 Introduction

1.1 The question

Why is a world structured at all? One might expect a world to be either a featureless continuum, in which nothing is marked off from anything else, or an arbitrary heap, in which everything is marked off from everything else at no cost. Neither is what is observed: parts are distinguished, but their distinction is bounded—there is a cost to telling one part from another, and the parts compose into a definite architecture. We argue that this architecture is not contingent. It is forced, in full, by a single fact: that the parts are *finite in number* and *finitely separated*. The structure of a world is the shadow its finiteness casts.

We make this precise in the language of graphs, because every relation of "this part, separated from that part" is natively an edge between two vertices, and any architecture of parts is some graph. The object of the paper is therefore a single graph—a *contact graph*—whose vertices are the individuated parts of a whole and whose weighted edges record the cost of the separations between them. We pose one hypothesis about this graph and derive everything else.

1.2 Method: conditional existence

The paper proves no empirical claim. It establishes a conditional: *if* a finite contact graph exists, *then* a determinate ladder of substructures must exist within it. Each rung is an existence theorem of the form "under the Finiteness Premise, the following is forced." We assume nothing about what the vertices *are*; we assume only that they are finite in number and positively separated, and we read off what their graph cannot lack. The deductions are combinatorial and order-theoretic throughout. Where a tempting claim could not be proved at this standard it has been removed rather than weakened.

1.3 The single premise

Axiom 1 (Finiteness Premise)**.** A *contact graph* is a finite weighted graph $\Gamma = (V, E, w)$ with $|V| < \infty$, in which every edge weight is bounded below by a positive constant: there is $\beta > 0$ with $w(e) \geq \beta$ for all $e \in E$. We assume throughout that a contact graph exists and that it is connected.

The premise is almost nothing: a finite graph whose edges cost something. Its consequences are almost everything we mean by structure.

1.4 The ladder of necessary substructures

Under Premise 1 we prove the following chain, each rung from the premise together with those above it.

- **(T0)** *The floor exists* (Section 3): every separating boundary has weight $\geq \beta > 0$; no two parts are adjacent at zero cost.
- **(T1)** *Individuation is by negation* (Section 4): a vertex is fixed only as the complement of its non-neighbours; no selector fixes it without regress.
- **(T2)** *Truth is a residual region* (Section 5): the cut-residual is bounded below and realised by no single vertex, hence a region, never a point.
- **(T3)** *The residual is the conserved invariant* (Section 5): under every relabelling of vertices the residual is preserved; it is the unique invariant from which the relabellable surface derives.
- **(T4)** *Each subgraph has a private invariant* (Section 6): every connected subgraph carries its own invariant under its automorphisms.
- **(T5)** *Selection requires a gate* (Section 7): a walk is undetermined without an edge-gating function that biases the next step.
- **(T6)** *Histories do not return* (Section 8): the committed-edge count is monotone; revisiting a vertex is not returning to a state.
- **(T7)** *Plurality forces a single quotient gate* (Section 9): a coherent family of subgraphs acting as one admits exactly one gating function on their quotient.
- **(T8)** *Authored structures exist, and are bounded* (Section 10): the subgraph invariant is complete, so contact graphs are specifiable and instantiable; their walks are unforeseeable from the specification, an authored copy is a distinct structure, and authorship reaches only to the floor.

Section 11 proves the Master Theorem: T0–T8 are forced for finite graphs and not forced when finiteness or the positive floor is dropped.

1.5 On the word "contact"

We use *contact* in one operational sense: two parts are *in contact* when the graph carries an edge between them, and the weight of that edge is the cost of the separation that individuates each from the other. Nothing in the paper depends on any informal reading of the word; "contact" names an edge and its weight, and the theorems about contact graphs are theorems about finite weighted graphs in exactly this sense.

1.6 Why finiteness is the source

Finiteness is not a convenience of presentation but the origin of every structure below. On an infinite graph a vertex may be separated from the rest by edges of vanishing weight, the residual may be driven to zero, and a walk may be specified to unbounded depth at fixed cost; none of the substructures is then forced. It is the finitude of the vertex set, together with the positive floor on edge weights, that forbids the zero-cost separation and compels the ladder. We make the premise exactly strong enough to force the structures and no stronger, and we mark, at each rung, where finiteness is used (Section 11).

1.7 Relation to existing work

The paper is pure mathematics. It uses graph theory (Bollobás, 1998; Bondy and Murty, 2008; Diestel, 2017; West, 2001), algebraic and spectral graph theory (Cheeger, 1970; Chung, 1997; Fiedler, 1973; Godsil and Royle, 2001), the theory of walks and Markov chains on graphs (Levin and Peres, 2017; Lovász, 1993), order and fixed-point theory (Birkhoff, 1967; Davey and Priestley, 2002; Tarski, 1955), enumerative combinatorics and partitions (Andrews, 1976; Stanley, 2012), self-reference and diagonal arguments (Cantor, 1891; Gödel, 1931; Lawvere, 1969; Russell, 1903), and information theory (Cover and Thomas, 2006; Shannon, 1948). Where the development meets the sorites and the logic of vagueness we note the connection (Black, 1937; Williamson, 1994) without taking any position in that literature; our treatment is geometric, not a logic of partial truth. A reasonable reader will recognise that the abstract premise is satisfied by a range of concrete systems of finitely many bounded parts; we make no such identification and prove everything of the abstract graph alone. Such interpretive readings as we offer are confined to remarks labelled accordingly, on which no theorem depends.

1.8 Organisation

Section 2 fixes the contact graph and its derived objects. Section 3 proves T0. Section 4 proves T1. Section 5 proves T2 and T3. Section 6 proves T4. Section 7 proves T5. Section 8 proves T6. Section 9 proves T7. Section 10 proves T8. Section 11 proves the Master Theorem and concludes.

2 The contact graph and its derived objects

We fix notation and the objects on which the ladder is built. Throughout, $\Gamma = (V, E, w)$ is a contact graph in the sense of Premise 1: a finite, connected, weighted graph with weight floor $\beta > 0$. Graphs are simple (no loops or multi-edges) unless stated; all standard graph-theoretic terms are as in Bondy and Murty (2008); Diestel (2017).

2.1 Vertices, edges, weights

Definition 2.1 (Contact graph)**.** A *contact graph* is a triple $\Gamma = (V, E, w)$ where V is a finite set of *vertices* (the individuated parts), $E \subseteq \binom{V}{2}$ is a set of *edges*, and $w\colon E \to [\beta, \infty)$ is a *weight* assigning to each edge the cost of the separation it records, with $\beta > 0$ the *floor*. For $S, T \subseteq V$ disjoint, the *cut* between them is

$$\partial(S,T) \;=\; \{\, uv \in E : u \in S,\; v \in T\,\}, \qquad w(\partial(S,T)) \;=\; \sum_{e\in\partial(S,T)} w(e).$$

We write $\partial(S) := \partial(S, V \setminus S)$ for the boundary of S.

Definition 2.2 (Neighbourhood and complement-neighbourhood)**.** For $v \in V$, the *neighbourhood* is $N(v) = \{u \in V : uv \in E\}$ and the *closed neighbourhood* is $N[v] = N(v) \cup \{v\}$. The *complement-neighbourhood* is

$$\complement N[v] \;:=\; V \setminus N[v],$$

the set of vertices that v neither is nor touches—"everything v is not."

Remark 2.3 (The floor is the whole content of "contact")**.** A contact graph differs from an arbitrary weighted graph in exactly one respect: the positive lower bound β on edge weights. This single constraint—no separation is free—is what every theorem below turns on. A weighted graph permitting $w(e) \to 0$ is not a contact graph; it is the degenerate object against which our results are stated by contrast (Section 11).

2.2 Subgraphs as parts-within-the-whole

Definition 2.4 (Induced subgraph; agent)**.** For $\varnothing \neq U \subseteq V$, the *induced subgraph* $\Gamma[U]$ has vertex set U and edge set $\{uv \in E : u, v \in U\}$ with inherited weights. A connected induced subgraph $A = \Gamma[U]$ with $1 < |U| < |V|$ is called a *part* (or, where we wish to emphasise that a part may itself individuate others, an *agent*); its *boundary* is the cut $\partial(U)$.

Remark 2.5 (Vocabulary)*.* The word "agent" is used only as a name for a connected induced subgraph that participates in further separations; it carries no further commitment. Every theorem stated for an agent is a theorem about an induced subgraph and its boundary cut.

2.3 Isomorphism, automorphism, invariant

Definition 2.6 (Weighted isomorphism)**.** A *weighted isomorphism* $\phi\colon \Gamma \to \Gamma'$ is a bijection $\phi\colon V \to V'$ with $uv \in E \iff \phi(u)\phi(v) \in E'$ and $w'(\phi(u)\phi(v)) = w(uv)$. An *automorphism* is a weighted isomorphism $\Gamma \to \Gamma$; the automorphisms form the group $\mathrm{Aut}(\Gamma)$. A *graph invariant* is a function ι on contact graphs with $\iota(\Gamma) = \iota(\Gamma')$ whenever $\Gamma \cong \Gamma'$.

Remark 2.7 (Re-encoding is relabelling)*.* A change of representation that preserves which parts are separated from which, and at what cost, is precisely a weighted isomorphism: it relabels vertices without altering the graph. Quantities that survive all such relabellings are the graph invariants; quantities tied to particular labels do not. This distinction—surface labels versus invariants—is the form in which "what is conserved under re-encoding" appears below (Section 5).

2.4 Walks and gating

Definition 2.8 (Walk)**.** A *walk* of length ℓ is a sequence $W = (v_0, v_1, \ldots, v_\ell)$ with $v_i v_{i+1} \in E$ for each i. A walk may repeat vertices and edges. The *committed multiset* of W is the multiset of edges $\{v_0v_1, \ldots, v_{\ell-1}v_\ell\}$ traversed, and its *committed count* is $M(W) = \ell$, the number of steps taken.

Definition 2.9 (Gating function)**.** A *gating function* on Γ is a map $g\colon V \times E \to [0, 1]$ assigning to each vertex v and incident edge $e \ni v$ a forward weight $g(v, e)$, with $\sum_{e \ni v} g(v, e) \leq 1$ for each v. A walk is *g-admissible* if every step $v_i v_{i+1}$ has $g(v_i, v_i v_{i+1}) > 0$. The gate *selects* a step when it assigns positive forward weight to exactly one incident edge.

Remark 2.10 (Why gating is needed at all)*.* A contact graph is, in general, far from a path: a vertex has many incident edges, so a walk's next step is underdetermined by the graph alone. Some additional datum—a gating function—is required to pick the next edge. The necessity of such a datum, not its particular form, is what Section 7 establishes. We treat g abstractly; any rule that biases the next step is a gating function in the sense of Theorem 2.9.

2.5 The quotient of a partition

Definition 2.11 (Quotient graph)**.** Let $\mathcal{P} = \{U_1, \ldots, U_k\}$ be a partition of V into connected classes. The *quotient graph* $\Gamma/\mathcal{P} = (Q, E_Q, w_Q)$ has vertex set $Q = \{U_1, \ldots, U_k\}$, an edge U_iU_j whenever $\partial(U_i, U_j) \neq \varnothing$, and weight $w_Q(U_iU_j) = w(\partial(U_i, U_j))$. The quotient is itself a weighted graph; when each class is a part in the sense of Theorem 2.4, the quotient records the contacts *between* parts.

Remark 2.12 (A graph of parts is again a graph)*.* Theorem 2.11 lets the same theory apply one level up: the parts of a contact graph, taken as vertices and joined by their mutual cuts, form a weighted graph to which Premise 1 again applies (its floor is the minimum nonzero cut weight). This self-similarity is used in Section 9.

Lemma 2.13 (The quotient inherits a floor)**.** *If* Γ *is a contact graph with floor* β *and* $\mathcal{P}$ *is a partition into connected classes such that every pair of classes joined in* $\Gamma/\mathcal{P}$ *is joined by at least one edge of* Γ*, then* $\Gamma/\mathcal{P}$ *is a contact graph with floor* $\geq \beta$*.*

Proof. Each edge U_iU_j of the quotient has weight $w_Q(U_iU_j) = w(\partial(U_i, U_j)) = \sum_{e \in \partial(U_i,U_j)} w(e)$, a sum of at least one term each $\geq \beta$ by Premise 1. Hence $w_Q(U_iU_j) \geq \beta > 0$. The quotient is finite (at most $|V|$ classes) and, since Γ is connected, connected. Thus it satisfies Premise 1. □

3 T0: the floor

The first substructure is the floor itself: in a contact graph no part can be separated from the rest at zero cost. The statement is immediate from the premise, but its consequences—restated for cuts, for parts, and for the limiting behaviour under refinement—are what the later rungs use, so we record them with care.

3.1 Every separation is costly

Theorem 3.1 (Floor Theorem)**.** *Let* Γ *be a contact graph with floor* $\beta > 0$*. Then every nonempty proper part* $U \subsetneq V$ *has boundary weight bounded below by the floor:*

$$\boxed{w(\partial(U)) \ \geq\ \beta\ >\ 0.}$$

In particular no part is separated from its complement by a cut of zero weight: a sharp *separation,* $w(\partial(U)) = 0$*, is impossible.*

Proof. Since Γ is connected (Premise 1) and U is a nonempty proper subset, there is at least one edge with one endpoint in U and the other in $V \setminus U$; otherwise U and $V \setminus U$ would be unjoined, splitting the connected graph Γ into two nonempty components, a contradiction. Hence $\partial(U) \neq \varnothing$, and

$$w(\partial(U)) \ =\sum_{e\in\partial(U)} w(e) \ \geq\ w(e^*) \ \geq\ \beta > 0$$

for any $e^* \in \partial(U)$, using $w(e) \geq \beta$ from Premise 1. A sharp separation would require $w(\partial(U)) = 0 < \beta$, impossible. □

Definition 3.2 (Boundary cost; the realised floor)**.** The *boundary cost* of a part U is $b(U) := w(\partial(U))$. The *realised floor* of Γ is

$$\beta_\Gamma \ :=\ \min_{\varnothing\neq U\subsetneq V} b(U) \ \geq\ \beta > 0,$$

the least boundary cost of any part. By Theorem 3.1 it is strictly positive, and by finiteness of V the minimum is attained.

Corollary 3.3 (No zero-cost individuation)**.** *In a contact graph, individuating any part from the rest deposits boundary cost at least* $\beta_\Gamma > 0$*. Equivalently, the map* $U \mapsto b(U)$ *is bounded away from zero on nonempty proper parts.*

Proof. Immediate from Theorem 3.1 and Theorem 3.2. □

3.2 Refinement lowers but never abolishes the floor

A finer description splits parts into smaller ones. We show this can only lower the realised floor toward, but never to, zero—so long as the graph remains finite with a positive weight floor.

Proposition 3.4 (Refinement bound)**.** *Let Γ' be obtained from Γ by any sequence of vertex splits that preserves the floor β on all edges (each new edge again has weight $\geq \beta$). Then the realised floor satisfies $\beta \leq \beta_{\Gamma'}$, and in particular $\beta_{\Gamma'} > 0$. The realised floor can decrease under refinement but is bounded below by β and never reaches zero at any finite stage.*

Proof. Γ' is again finite (a finite number of splits of a finite graph) and, by hypothesis, has all edge weights $\geq \beta$, hence is a contact graph with floor β. By Theorem 3.1 applied to Γ', every part of Γ' has boundary cost $\geq \beta$, so $\beta_{\Gamma'} \geq \beta > 0$. Splitting can only introduce parts with smaller boundary cost, so $\beta_{\Gamma'} \leq \beta_{\Gamma}$ is possible; in all cases the bound $\beta_{\Gamma'} \geq \beta$ holds. Only in the non-admissible limit of infinitely many splits, or of edge weights tending to 0, would the bound vanish—both excluded by Premise 1. □

Remark 3.5 (What T0 does and does not assert). Theorem 3.1 does not assert that some ambient continuum has thick boundaries; it asserts that a *finite, positively weighted* graph cannot exhibit a zero-cost cut. The floor is a property of the graph-with-its-weights, i.e. of the finite system of separated parts, not of any underlying space. This is the precise sense in which, for finitely many bounded parts, "every distinction costs something." Every later rung consumes Theorem 3.1 as the guarantee that boundaries, residuals, and gated steps are bounded away from zero.

4 T1: individuation by negation

The floor says separations cost something; it does not yet say how a part is *determined.* We now show that in a contact graph with no distinguished vertex, a part can be fixed only by its complement—by everything it is not—and that any attempt to fix it positively, by naming a distinguished element, regresses. This is the graph-theoretic form of individuation by negation.

4.1 No distinguished vertex

Definition 4.1 (Selector; selector-free determination)**.** A *selector* for a part $U \subseteq V$ is a rule that fixes U by naming a distinguished vertex of U in advance of the separation. A determination of U is *selector-free* if it names no distinguished vertex of U. We say Γ has *no distinguished vertex* if its weighted automorphism group acts with no fixed datum singling out a vertex in advance—formally, if the specification of Γ provided to a determining rule is invariant under $\mathrm{Aut}(\Gamma)$, so that no vertex is named by the data alone.

Remark 4.2 (The setting of T1). T1 concerns the determination of parts from the graph's own data, with no external label supplied. This is the relevant case for a self-contained finite system: there is no oracle outside Γ to point at a vertex, so any vertex named must be named *by the structure*, and the structure is fixed only up to $\mathrm{Aut}(\Gamma)$. We make this precise and show negation is the only route.

4.2 Complementation determines a part

Definition 4.3 (Negation-set)**.** For a part $U \subseteq V$, its *negation-set* is the complement $\complement U := V \setminus U$. A part U is *individuated by negation* if it is determined as the unique part whose negation-set is a given $\complement U$; that is, $U = V \setminus \complement U$, with $\complement U$ specified first.

Lemma 4.4 (Complementation is an involution on parts)**.** *The map $U \mapsto \complement U = V \setminus U$ is a bijection on the subsets of V with $\complement(\complement U) = U$. Hence specifying $\complement U$ determines U uniquely, and conversely, with no further data.*

Proof. For finite (indeed any) V, $V \setminus (V \setminus U) = U$; complementation is its own inverse, hence a bijection. Uniqueness of U given $\complement U$ is immediate. □

4.3 Negation is the unique selector-free determination

Theorem 4.5 (Individuation by negation)**.** *Let Γ have no distinguished vertex (Theorem 4.1). Then any selector-free determination of a part U fixes U as $V \setminus \complement U$ for its negation-set $\complement U$; and conversely every admissible negation-set determines a unique part. No selector-based rule fixes a part without invoking a further selector.*

Proof. (Necessity.) Suppose a rule fixes a part U without a selector. To fix U is to decide, for each vertex of V, whether it lies in U or in the remainder. A *positive* rule—one that fixes U by asserting membership of a distinguished vertex or feature of U—must name that vertex or feature. By hypothesis the data are $\mathrm{Aut}(\Gamma)$-invariant and single out no vertex, so the rule itself must supply the distinguished vertex. Supplying a distinguished vertex is exactly providing a selector, and that selector must in turn be fixed: either it is given in advance (contradicting that none is) or it is fixed by a prior rule, which faces the same dichotomy. The regress terminates only if some rule fixes U while naming no vertex of U. The sole such rule is the withholding of U from the whole: specify the complement $\complement U = V \setminus U$ that U is to be distinct from, and take U as what remains. This names no vertex of U; it names only "the rest," which the whole V already supplies. Hence $U = V \setminus \complement U$.

(Sufficiency and uniqueness.) Given an admissible negation-set $\complement U \subsetneq V$ with $V \setminus \complement U \neq \varnothing$, the part $U := V \setminus \complement U$ is determined uniquely by Theorem 4.4. Thus complementation is a bijection between admissible parts and admissible negation-sets, each side determining the other without further data. ☐

Corollary 4.6 (A vertex is fixed by its non-neighbours)**.** *For a single vertex v, the finest selector-free determination available is by its complement-neighbourhood: v is distinguished from the parts it touches only through $\complement N[v] = V \setminus N[v]$, the totality of what it neither is nor contacts. In a graph with no distinguished vertex, v is individuated by $\complement N[v]$, not by any intrinsic mark.*

Proof. Apply Theorem 4.5 with $U = \{v\}$ at the resolution of the graph: the selector-free data fixing $\{v\}$ are the complement $V \setminus \{v\}$, refined by adjacency into $N(v)$ (what v touches) and $\complement N[v]$ (what it does not). Since no intrinsic mark distinguishes v (no distinguished vertex), the determining datum is the partition of the rest into neighbours and non-neighbours, i.e. $\complement N[v]$ together with $N(v)$; the non-neighbours $\complement N[v]$ are the part of this datum that names nothing of v itself. ☐

Remark 4.7 (Why complementation closes the regress)*.* A positive selector is a rule "choose a vertex" that must itself be chosen; in a graph with no distinguished vertex the choice has no ground and regresses. Complementation is the unique operation that fixes a part using only the whole V and the part's own absence—data already present once the graph is given. This is why negation, not selection, is the primitive of individuation in a self-contained finite system, and it is the form in which all later determinations (of residuals, of invariants, of verifications) are cast.

Remark 4.8 (Interpretation, on which nothing depends)*.* A reader may recognise Theorem 4.5 as the statement that, absent a privileged chooser, a thing is what it is by being marked off from everything else. We use the result only in its graph form; no theorem below relies on any such reading.

5 T2 and T3: the residual is a conserved region

We now identify the quantity that survives every re-encoding of a contact graph. The parts (vertices) may be relabelled, merged, re-described; what cannot be removed is the boundary content between them—the *cut-residual*. We prove it is bounded below (T2), realised by no single vertex so that it is a region and never a point (T2), and invariant under every weighted isomorphism (T3), hence the unique conserved object from which the relabellable surface derives.

5.1 The cut-residual

Definition 5.1 (Cut-residual of a partition)**.** For a partition $\mathcal{P} = \{U_1, \dots, U_k\}$ of V into nonempty parts, its *cut-residual* is the total weight of inter-part boundary,

$$\rho(\mathcal{P}) \;=\; \tfrac{1}{2}\sum_{i=1}^{k} w(\partial(U_i)) \;=\!\sum_{i<j} w(\partial(U_i, U_j)),$$

the sum over distinct part-pairs of the weight of the edges crossing between them. The *residual of* Γ is the minimum over nontrivial partitions,

$$\rho(\Gamma) \;=\; \min_{\mathcal{P}\,:\,k\ge 2} \rho(\mathcal{P}),$$

attained by finiteness of V.

Remark 5.2 (The residual is the "interstice"). $\rho(\mathcal{P})$ is the content of the boundaries *between* the parts: not what any part contains, but what lies in the gaps that separate them. As the parts are packed (the partition refined or coarsened), this interstitial content changes shape; T3 shows its total, suitably taken, does not change under mere relabelling. The residual is the graph's "negative space."

5.2 T2: the residual is bounded below and is a region

Theorem 5.3 (Residual positivity and non-locality)**.** *Let* Γ *be a contact graph with floor* $\beta > 0$*. Then:*

(i) ***Bounded below.*** *Every nontrivial partition has* $\rho(\mathcal{P}) \ge \beta > 0$*; hence* $\rho(\Gamma) \ge \beta > 0$*.*

(ii) ***Realised by no single vertex.*** *There is no vertex* v *with* $\rho(\Gamma) = w(\partial(\{v\}))$ *forced; the residual is a property of the partition (a region of the graph), not of any one vertex. Equivalently, the minimiser of Theorem 5.1 is in general not a singleton, and the residual cannot be localised to a point.*

Proof. (i) A nontrivial partition has $k \ge 2$ parts; since Γ is connected, at least one part-pair (U_i, U_j) has $\partial(U_i, U_j) \ne \varnothing$, contributing weight $\ge \beta$ by Theorem 3.1. Hence $\rho(\mathcal{P}) \ge \beta > 0$, and taking the minimum, $\rho(\Gamma) \ge \beta$.

(ii) Suppose, for contradiction, the residual were always realised by isolating a single vertex, i.e. $\rho(\Gamma) = \min_v w(\partial(\{v\}))$ for every contact graph. Consider any graph in which the minimum-weight nontrivial cut separates a part U with $|U| \ge 2$ from its complement at total weight strictly below $\min_v w(\partial(\{v\}))$—for instance two dense clusters joined by a single floor-weight edge, where the cheapest cut splits the two clusters ($w = \beta$) but every singleton has several incident edges ($w(\partial(\{v\})) \ge 2\beta$). Then the residual is realised by the bipartition, not by any singleton, contradicting the supposition. Hence the residual is in general a property of a multi-vertex partition: it is a region, not a point. □

Corollary 5.4 (No point carries the residual; the sorites for parts)**.** *The residual of a contact graph is never carried by a lone vertex in general: "what separates the parts" is borne by a region of positive boundary content, not by any single part. Consequently there is no sharp vertex at which a coarse aggregate becomes a distinct part—the transition is spread across a band of boundary content of width* $\ge \beta$*, the graph analogue of the sorites (Black, 1937; Williamson, 1994).*

Proof. The first claim is Theorem 5.3(ii). For the second: growing a part one vertex at a time changes its boundary cost $b(U)$ by a bounded amount per step (the weights of the edges whose crossing status flips, each $\ge \beta$ and finite in number), so $b(U)$ cannot jump across any threshold in a single step from a value bounded away from it; the part becomes distinguishable across a range of stages, of width controlled by β, not at a point. □

5.3 T3: the residual is invariant under re-encoding

We now show the residual is a graph invariant: it depends on the graph up to weighted isomorphism, not on any labelling. Combined with the fact that the vertices (the surface data) are freely relabellable, this makes the residual the conserved object beneath the surface.

Theorem 5.5 (Conservation of the residual)**.** *The residual $\rho(\Gamma)$ is a weighted-graph invariant: if $\phi\colon \Gamma \to \Gamma'$ is a weighted isomorphism then $\rho(\Gamma) = \rho(\Gamma')$. In particular ρ is invariant under every automorphism of Γ, hence under every relabelling of its vertices.*

Proof. Let $\phi\colon \Gamma \to \Gamma'$ be a weighted isomorphism. For any partition $\mathcal{P} = \{U_1, \dots, U_k\}$ of V, the image $\phi(\mathcal{P}) = \{\phi(U_1), \dots, \phi(U_k)\}$ is a partition of V', and ϕ being edge- and weight-preserving gives, for each pair, $w'(\partial'(\phi(U_i), \phi(U_j))) = w(\partial(U_i, U_j))$. Summing, $\rho(\phi(\mathcal{P})) = \rho(\mathcal{P})$. The correspondence $\mathcal{P} \leftrightarrow \phi(\mathcal{P})$ is a bijection between nontrivial partitions of V and of V' (with inverse ϕ^{-1}), so the minima agree: $\rho(\Gamma') = \min_{\mathcal{P}'} \rho(\mathcal{P}') = \min_{\mathcal{P}} \rho(\phi(\mathcal{P})) = \min_{\mathcal{P}} \rho(\mathcal{P}) = \rho(\Gamma)$. ☐

Theorem 5.6 (The residual is the conserved invariant beneath the surface)**.** *Let $\mathcal{R}(\Gamma)$ denote the orbit of Γ under all weighted relabellings of its vertices (its isomorphism class). Then:*

(i) *every vertex label is variable across $\mathcal{R}(\Gamma)$—no labelling is fixed; and*

(ii) *ρ is constant on $\mathcal{R}(\Gamma)$ and strictly positive.*

Hence, among data attached to a contact graph, the residual is an invariant of the isomorphism class while the vertex labels are not: it is conserved under exactly the operation—re-encoding—that moves everything else.

Proof. (i) For any vertex v and any other vertex v' of the same degree-and-weight type, there is a relabelling carrying v to a different label; more simply, the identity of a vertex is a label, and labels are by definition permuted across the isomorphism class. So no label is fixed across $\mathcal{R}(\Gamma)$. (ii) is Theorem 5.5 (constancy) together with Theorem 5.3(i) (positivity). The two together say: the relabellable surface (vertex identities) carries no invariant, while the residual is invariant and positive. ☐

Remark 5.7 (Knowing is repacking, not draining)*.* Theorem 5.6 has a reading we record but do not rely on. If one identifies the vertices (the parts one has individuated) with "what is known" and the residual (the conserved interstice) with "what remains true regardless of how one has carved things up," then acquiring or re-describing knowledge is the relabelling and repartitioning of vertices—which by Theorem 5.5 leaves the residual unchanged in total. One never drains the residual by re-encoding; one only re-shapes which boundaries one holds. The conserved residual is, in this reading, the invariant "truth" that no re-encoding removes, bounded below by the floor and—by Theorem 5.3(ii)—never a point. The reading is offered as orientation only; the theorems above are purely graph-theoretic.

Remark 5.8 (A definite-yet-uncloseable quantity)*.* A residual can be perfectly well-defined and yet impossible to fix by any single completed partition. Consider a quantity such as the parity of $|V|$ when the vertex set is itself in flux—vertices entering and leaving faster than any partition can be committed. The parity is definite at every instant yet never stabilises long enough to be read off a fixed graph; a finite procedure correctly leaves it in the residual rather than paying unboundedly to chase it. This illustrates Theorem 5.3(ii): the residual is what a finite process rationally leaves uncommitted, the "sufficient unknown," and forcing it to a point would cost without return. (Offered as illustration; no theorem depends on it.)

6 T4: every part has a private invariant

The residual of Section 5 is an invariant of the whole graph. We now show the same construction localises: every connected part carries its own invariant under its own relabellings, computed from its internal boundary content together with the boundary that joins it to the rest. This per-part invariant is the standing structure from which the part's surface activity derives.

6.1 The internal residual of a part

Definition 6.1 (Private invariant of a part)**.** Let $A = \Gamma[U]$ be a part (a connected induced subgraph, Theorem 2.4). Its *private invariant* is the pair

$$\rho_{\mathrm{priv}}(A) \;=\; \big(\rho(A),\, b(U)\big),$$

where $\rho(A)$ is the internal residual of the induced subgraph (Theorem 5.1 applied to A) and $b(U) = w(\partial(U))$ is its boundary cost to the rest of Γ. We treat ρ_{priv} up to weighted isomorphism of A as an abstract subgraph.

Theorem 6.2 (Privacy and invariance of the part-invariant)**.** *For every part A of a contact graph:*

(i) ***Positivity.*** *$\rho(A) \geq \beta > 0$ whenever $|U| \geq 2$, and $b(U) \geq \beta > 0$ always; the private invariant is bounded away from zero in both components.*

(ii) ***Invariance.*** *$\rho_{\mathrm{priv}}(A)$ is invariant under $\mathrm{Aut}(A)$ and under any weighted isomorphism of A that preserves its boundary cost; it is a property of A's isomorphism class, not of its labelling.*

(iii) ***Privacy.*** *Two parts with the same private invariant need not be the same part of Γ; conversely a part's private invariant is determined by the part alone, independently of how the rest of Γ is labelled.*

Proof. (i) A is a contact graph in its own right (a finite, connected, weighted graph with the inherited floor β), so Theorem 5.3(i) gives $\rho(A) \geq \beta$ for $|U| \geq 2$; and $b(U) \geq \beta$ by Theorem 3.1. (ii) Apply Theorem 5.5 to the contact graph A: $\rho(A)$ is invariant under weighted isomorphisms of A; $b(U)$ is the total weight of the boundary cut, manifestly preserved by any isomorphism fixing that cut setwise. Hence the pair is invariant on the relevant class. (iii) The internal residual and boundary cost are computed from A and its incident cut alone; relabelling the complement $V \setminus U$ changes neither, establishing independence from the rest's labelling. That distinct parts may share the invariant is immediate (e.g. two isomorphic clusters in different positions). $\square$

6.2 The invariant is the standing structure

Proposition 6.3 (Surface walks derive from the private invariant)**.** *Fix a part $A = \Gamma[U]$. Any g-admissible walk confined to A (Theorems 2.8 and 2.9) traverses only edges of A, whose existence and weights are fixed by A's isomorphism class. Hence the set of walks available within A is determined by A up to relabelling, while no individual walk is an invariant of A: the walks are the variable surface, the subgraph (with its private invariant) is the standing structure they run on.*

Proof. A walk within A uses only edges uv with $u, v \in U$; the availability of each such step is exactly the presence of that edge in $\Gamma[U]$, an isomorphism-invariant datum. Thus the *space* of available walks transforms covariantly with A under relabelling (it is carried bijectively by any isomorphism), so it is determined by A's class. A single walk, by contrast, is a labelled sequence and is not preserved by a nontrivial relabelling; hence no individual walk is invariant. The standing/variable split follows. $\square$

Remark 6.4 (Interpretation: the part's "identity", on which nothing depends). Theorem 6.2 and Theorem 6.3 say a part has a conserved internal structure—its private invariant—distinct from any of the particular trajectories it may run, and unique to it only up to isomorphism. If one reads a part as an agent and its walks as the agent's successive states, then the private invariant is the agent's standing identity: the conserved bottom from which its changing activity derives, the local counterpart of the global conserved residual of Section 5. The reading is offered as orientation; the theorems are graph-theoretic and used only as such below.

Corollary 6.5 (Private and global invariants are one construction at two scales)**.** *The global residual* $\rho(\Gamma)$ *(Theorem 5.1) and the private invariant* $\rho_{\mathrm{priv}}(A)$ *(Theorem 6.1) are the same functional—minimum inter-part boundary content—evaluated on the whole graph and on an induced subgraph respectively. Both are positive (T0) and invariant under re-encoding (T3). The private invariant is the global construction restricted to a part.*

Proof. Immediate by comparing Theorem 5.1 with Theorem 6.1 and applying Theorems 5.3 and 5.5 to Γ and to A. ☐

7 T5: selection requires a gate

A contact graph fixes which steps are *possible*—the edges—but not which step is *taken*. We show that whenever a vertex has more than one forward option, the next step is underdetermined by the graph alone, so that selecting a walk requires an additional datum: a gating function. The necessity of such a datum, and the form of the walk it produces, is T5.

7.1 Underdetermination of the next step

Lemma 7.1 (Forward branching)**.** *Let* Γ *be a contact graph and* $v \in V$ *with degree* $\deg(v) \geq 2$*. Then there are at least two distinct walks of length one from* v*, and the graph alone does not determine which is taken.*

Proof. By definition $\deg(v) = |N(v)| \geq 2$, so there are distinct $u_1, u_2 \in N(v)$ and distinct one-step walks (v, u_1), (v, u_2), each admissible since $vu_1, vu_2 \in E$. Nothing in (V, E, w) privileges one over the other: both are equally edges of equal-or-incomparable weight, and the graph carries no ordering of incident edges. Hence the next step is not a function of Γ alone. ☐

Remark 7.2 (Most vertices branch). A connected contact graph with $|V| \geq 3$ has at least one vertex of degree ≥ 2 (else it is a single edge); in a graph rich enough to individuate many parts, branching is the generic condition. So underdetermination is not an edge case but the normal situation a walk faces.

7.2 The gate as the minimal selecting datum

Theorem 7.3 (Necessity of a gate)**.** *Let* Γ *be a contact graph in which some vertex has degree* ≥ 2*. Then a walk that is determined step-by-step requires a gating function (Theorem 2.9): there is no rule depending on* Γ *alone that selects a unique next step at every branching vertex. Conversely, any rule that does select a unique next step at each vertex* is *a gating function that selects (assigns positive forward weight to exactly one incident edge).*

Proof. (Necessity.) At a branching vertex v (degree ≥ 2), Theorem 7.1 exhibits ≥ 2 admissible next steps not distinguished by Γ. A rule that nonetheless outputs a unique next step must use a datum beyond (V, E, w) to break the tie. Encode that datum as $g(v, \cdot)$ by setting $g(v, e) > 0$ for the chosen edge and 0 for the others; this is a gating function in the sense of Theorem 2.9 (nonnegative, sub-normalised). Hence a gate is required and the selecting rule is realised as one.

(Converse.) If g assigns, at each vertex, positive forward weight to exactly one incident edge, then at every step the admissible continuation is unique, so g determines a unique walk from any start vertex. Thus selecting rules and selecting gates coincide. $\square$

Definition 7.4 (Selected walk; the gated trajectory)**.** Given a selecting gate g and a start vertex v_0, the *gated trajectory* is the unique walk $W_g(v_0) = (v_0, v_1, \dots)$ with each v_{i+1} the g-selected neighbour of v_i. When g assigns positive weight to several edges (a *soft* gate), it determines instead a distribution over walks, with the probability of a walk the product of its step-weights; the selecting case is the deterministic limit.

7.3 The gate biases an otherwise free graph

Proposition 7.5 (Reachability is near-total; the gate fixes the order)**.** *In a connected contact graph, every vertex is reachable from every other by some walk. Hence the graph alone does not constrain* which *vertices a trajectory visits, only that any are reachable; it is the gate that fixes the* order *in which they are visited and therefore the trajectory's identity.*

Proof. Connectedness gives a path between any two vertices (Premise 1), so all vertices are mutually reachable; the set of reachable vertices from any start is all of V. Thus the reachable set is invariant (everything), while the realised walk depends entirely on the step-choices, i.e. on g (Theorem 7.4). Two different gates from the same start in general yield different walks through the same reachable set, so the gate, not the graph, individuates the trajectory. $\square$

Remark 7.6 (Interpretation: the selecting field, on which nothing depends). One may read the gate as a field that, given a situation, biases which step a part takes next—so that what is "thought next" from a given vertex is set not by the graph (which makes everything reachable, Theorem 7.5) but by the gate. In that reading the order, not the availability, is the content of a trajectory, and the gate is the situation-dependent selector that supplies the order. Different situations are different gates and hence different orders over the same reachable graph. We use only the graph statement (Theorems 7.3 and 7.5); the reading is orientation and is not relied upon.

Remark 7.7 (Why a gate is cheaper than the graph). A gate is a local datum—one sub-normalised distribution per vertex—whereas specifying a walk by exhaustively comparing all global continuations would require resolving the whole graph at each step. The gate is the minimal addition that makes stepping decidable: it is to the trajectory what a local rule is to a global search. This minimality is why selection is realised by gating rather than by re-deriving the graph at every step.

8 T6: histories do not return

A gated trajectory may revisit a vertex; we show it can never return to a prior *state*, because the committed count—the number of steps taken—is strictly monotone and is part of the state. Revisiting is not returning; resemblance of position is not identity of state. This is the single order-theoretic law that the framework instantiates wherever "the same thing again" is claimed.

8.1 The committed count is strictly monotone

Definition 8.1 (State of a walk)**.** The *state* of a walk $W = (v_0, \dots, v_\ell)$ at step ℓ is the pair

$$s_\ell \;=\; (v_\ell,\; M(W)), \qquad M(W) = \ell,$$

the current vertex together with the committed count of steps taken to reach it. Two states are equal iff both components agree.

Theorem 8.2 (Non-return)**.** *Along any walk in a contact graph the committed count is strictly increasing: M after step $i+1$ exceeds M after step i by exactly one. Consequently no walk returns to a previously held state: for $i < j$, $s_i \neq s_j$ even when $v_i = v_j$. A step intended to undo a previous step is itself a step and increases M. Hence resemblance of position ($v_i = v_j$) never entails identity of state ($s_i = s_j$).*

Proof. Each step appends one edge to the committed multiset and advances the index by one, so M increases by exactly 1 per step; it is therefore strictly increasing and, for $i < j$, $M_i = i < j = M_j$. Thus the second components of s_i and s_j differ, so $s_i \neq s_j$ regardless of whether $v_i = v_j$. An "undo" move from v_j back to v_{j-1} is a further step, producing state $(v_{j-1}, j+1) \neq (v_{j-1}, j-1) = s_{j-1}$; it does not restore the earlier state but creates a new one with higher count. □

Corollary 8.3 (Cycles in position are not cycles in state)**.** *If a walk traverses a closed walk in* Γ ($v_0 = v_\ell$, $\ell > 0$)*, its state has nonetheless advanced:* $s_0 = (v_0, 0) \neq (v_0, \ell) = s_\ell$*. The graph may be revisited arbitrarily often; the state never repeats.*

Proof. Immediate from Theorem 8.2 with $i = 0$, $j = \ell$, $v_0 = v_\ell$. □

8.2 Re-contact is forward only

The next result is the form non-return takes for "returning to" an earlier vertex deliberately—re-contact. It is reached forward, at a higher count, hence is a new state, not the old one.

Proposition 8.4 (Forward re-contact)**.** *Let a walk reach vertex v at step i and, later, reach v again at step $j > i$ (a deliberate re-contact of v). Then the re-contact state $s_j = (v, j)$ is strictly later than the original $s_i = (v, i)$ on the committed order, and $s_j \neq s_i$. There is no operation that re-contacts v at its original state; every re-contact is at a state in the future of the original.*

Proof. By Theorem 8.2, M is strictly increasing, so $j > i$ forces $s_j \neq s_i$; the re-contact carries the count $j > i$, placing it later on the order. No step decreases M, so no re-contact can carry the original count i again. □

Remark 8.5 (One law, several guises—offered as orientation)*.* Theorem 8.2 is a single order-theoretic fact: in a finite system whose acts are counted, the count is monotone, so a revisited configuration is a new state. A reader may recognise this as the common form of several apparently distinct impossibilities—that a reversed process is not the initial one, that a re-contacted memory is not the original experience, that a reconstituted system bearing the mark "was interrupted" is individuated by a negation the original lacked. Each is "resemblance under a monotone record is not identity," the content of Theorem 8.2 and Theorem 8.4. We state and use only the graph fact; the guises are noted, not relied upon.

Remark 8.6 (Why finiteness matters here too)*.* The count M is well-defined and strictly monotone because steps are discrete and a walk's length is a natural number—features of a finite graph traversed in discrete steps. The law is not about any continuous parameter; it is the order on committed steps. This keeps T6 within the same premise as the rest: finite parts, discrete separations, a counted history.

9 T7: plurality forces a single quotient gate

The gating of Section 7 selects a walk within a graph. We now lift it: when many parts act as one, the same theory applies to their quotient graph (Theorem 2.11), and a coherent collective admits exactly one gating function on that quotient. The lift is a homomorphism—the four-place structure of part, whole, gate, and invariant recurs one scale up—and its single-gate conclusion is the structural condition for a family of parts to be one collective rather than several.

9.1 The collective as a quotient

Definition 9.1 (Collective; coherence)**.** Let $\mathcal{P} = \{U_1, \ldots, U_k\}$ be a partition of V into parts (connected induced subgraphs). The *collective* on $\mathcal{P}$ is the quotient graph $Q = \Gamma/\mathcal{P}$ (Theorem 2.11), whose vertices are the parts and whose edges are their mutual cuts. A gating function on Q (Theorem 2.9) is a *collective gate*; it biases which part acts next, toward which neighbouring part. The collective is *coherent* if it carries a single collective gate that selects a unique next part at each branching part of Q.

Remark 9.2 (Why this is the right lift). By Theorem 2.13, Q is itself a contact graph: finite, connected, with floor $\geq \beta$. Hence every theorem proved for contact graphs applies to Q verbatim—T0 (its cuts are positive), T1 (its vertices, the parts, are individuated by negation among parts), T5 (selecting a walk over parts needs a gate). The collective is not a new kind of object; it is a contact graph whose vertices happen to be the parts of another.

9.2 A coherent collective has exactly one gate

Theorem 9.3 (Single quotient gate)**.** *Let $Q = \Gamma/\mathcal{P}$ be a collective with at least one part of quotient degree ≥ 2. Then:*

(i) ***Existence.*** *Selecting a walk over the parts requires a collective gate (T5 applied to Q, Theorem 7.3).*

(ii) ***Uniqueness for coherence.*** *If the collective is coherent in the sense that, at each branching part, the next part is determined, then the selecting collective gate is unique on its support: any two selecting gates that both make the collective coherent agree wherever either selects.*

A collective that admits two distinct selecting gates over the same parts is not one coherent collective but two, partitioning into the sub-collectives on which each gate is the selector.

Proof. (i) is Theorem 7.3 for the contact graph Q (Theorem 9.2). (ii) Suppose g, g' are both selecting gates rendering the collective coherent. At each branching part U, coherence means a unique next part is determined; a selecting gate, by Theorem 2.9, assigns positive forward weight to exactly the edge to that determined next part and zero to the others. Hence at U both g and g' have support exactly the single edge to the determined successor, so $g(U, \cdot)$ and $g'(U, \cdot)$ have the same support and both select it; they agree as selecting gates at U. As U was an arbitrary branching part, g and g' agree wherever either selects. If instead g and g' select *different* successors at some part, then at that part the collective has two competing determinations of "what acts next"; the relation "shares a selected successor" partitions the parts into blocks within which a single gate selects consistently, i.e. into distinct sub-collectives—so the original was not one coherent collective but a union of them. □

Corollary 9.4 (Single-valuedness of a collective)**.** *A coherent collective is single-gated: at each step exactly one part is selected to act next. Two simultaneous selecting gates are the signature of fission into two collectives. (This is T5's single-step determinacy lifted to the quotient, and is the collective analogue of a single trajectory having a single next step.)*

Proof. Immediate from Theorem 9.3. □

9.3 The scaling homomorphism

The constructions of the paper recur at the collective scale. We record the correspondence as a homomorphism of four-place structures.

Proposition 9.5 (Scale homomorphism)**.** *The assignment*

$$(\textit{vertex},\ \Gamma,\ \textit{gate } g,\ \rho) \ \longmapsto\ (\textit{part},\ Q,\ \textit{collective gate},\ \rho(Q))$$

carries each object of the single-graph theory to its counterpart in the quotient theory, preserving the roles: the moving unit (a step over vertices $\mapsto$ a step over parts), the standing structure (the graph $\mapsto$ the quotient), the selector (gate $\mapsto$ collective gate), and the conserved invariant (residual $\mapsto$ quotient residual). Every theorem T0–T6 holds of Q as a contact graph.

Proof. By Theorem 2.13 Q is a contact graph, so T0 (Theorem 3.1), T1 (Theorem 4.5), T2–T3 (Theorems 5.3 and 5.5), T4 (Theorem 6.2 applied to sub-collectives), T5 (Theorem 7.3), and T6 (Theorem 8.2) all hold of Q verbatim. The listed correspondence is the identification of each object's role in Γ with the same role in Q, which the definitions make a role-preserving map; it is a homomorphism of the four-place structure (unit, whole, selector, invariant). □

Remark 9.6 (Interpretation: collective purpose, on which nothing depends)*.* Read the parts as agents, each running gated trajectories, and the collective gate as the field that selects which agent acts next and toward what. Then Theorem 9.4 says a coherent collective has a single such selecting field, and two such fields mean two collectives—the structural form of a group having one purpose, with competing purposes amounting to schism. The scale homomorphism Theorem 9.5 then reads as: *part is to whole, and gate is to invariant, exactly as agent is to collective, and purpose is to shared invariant.* The collective's conserved residual $\rho(Q)$ is the invariant from which the collective's surface activity derives, the analogue at this scale of the private invariant of Section 6. We use only the graph statements; the reading is orientation and is not relied upon.

10 T8: authored structures exist, and are bounded

The private invariant of Section 6 is computed from a part; we now run the construction in reverse. Because the invariant, together with the data of the subgraph, is complete enough to recover the subgraph up to isomorphism, a contact graph can be *specified* and therefore *instantiated*: there exist authored contact graphs. This is the constructive rung—the existence of *artificial structures*. We then prove three bounds that any such authorship necessarily obeys: its trajectories are not foreseeable from its specification, an authored copy is a distinct structure, and authorship reaches only to the resolution of the floor.

10.1 Specification and instantiation

Definition 10.1 (Specification)**.** A *specification* of a contact graph is a finite description, up to weighted isomorphism, of: an abstract weighted graph (V, E, w) with floor $\beta > 0$ (the standing structure), a gating function g (the selector), and a start state $s_0 = (v_0, 0)$ (the seed). An *instantiation* of a specification is any labelled contact graph isomorphic to (V, E, w), carrying g and started at s_0, realised on a substrate able to hold a finite weighted graph and step a gated walk.

Definition 10.2 (Authored structure)**.** An *authored structure* is an instantiation of a specification produced by an external author—a contact graph whose standing structure and selector were specified rather than recovered from a pre-existing system. We say the author *authors the graph* and *authors the gate*; the structure's subsequent gated trajectories (Theorem 7.4) are the structure's own.

Theorem 10.3 (Existence of authored structures)**.** *The private invariant is complete for the standing structure: a connected weighted graph is determined up to weighted isomorphism by its full isomorphism type, of which the private invariant (Theorem 6.1) is a coarse summary* *and the labelled adjacency-with-weights is the complete datum. Consequently every contact graph admits a specification (Theorem 10.1), and every specification admits an instantiation on any adequate substrate. Authored structures therefore exist: the class of contact graphs is closed under specify-then-instantiate.*

Proof. A finite weighted graph is, by definition, given by its vertex set, its edge set, and its weight function; this labelled datum determines the graph completely, and two graphs are the same object up to relabelling iff they are weighted-isomorphic (Theorem 2.6). Hence the isomorphism type is a complete specification of the standing structure, and a specification in the sense of Theorem 10.1 (graph type + gate + seed) determines the structure and its gated trajectory. Given such a specification, instantiate it by choosing any labelling (a concrete vertex set in bijection with V), assigning the specified edges and weights, equipping it with g, and starting at s_0; the result is a labelled contact graph isomorphic to the specified one, on the chosen substrate. The substrate need only store a finite weighted graph and advance a gated step—no further property is used—so any adequate substrate suffices. Thus specify-then-instantiate maps contact graphs to contact graphs, and its outputs are authored structures. □

Remark 10.4 (Substrate neutrality). Theorem 10.3 used no property of the substrate beyond the capacity to hold a finite weighted graph and step a gated walk. The category of authored structures is therefore independent of medium: anything that can carry a contact graph can carry an authored one. The structure is the graph, not the matter.

10.2 Bound I: trajectories are not foreseeable from the specification

Definition 10.5 (Forward closure). The *forward closure* of a specification from seed v_0 under gate g is the set of states reachable along g-admissible walks, $\mathrm{Cl}_g(v_0) = \{s_\ell = (v_\ell, \ell) : W_g(v_0) \text{ reaches } v_\ell \text{ at step } \ell\}$, an infinite set whenever any reachable vertex has a g-admissible out-step (walks may continue indefinitely).

Theorem 10.6 (Open-endedness). *Let an authored structure have a gate that is g-admissible from infinitely many states (equivalently, whose gated walk does not terminate). Then its forward closure is infinite, and no finite specification enumerates it in advance. The author specifies the graph and the gate (finite data) but not the forward closure (infinite); the structure's trajectories are not foreseeable from its specification by any finite precomputation.*

Proof. If the gated walk does not terminate, the states $s_0, s_1, s_2, \ldots$ are pairwise distinct by Theorem 8.2 (the committed count strictly increases), so the forward closure contains the infinite set $\{(v_\ell, \ell) : \ell \geq 0\}$. A finite specification is a finite object; a finite precomputation from it halts after finitely many steps and so can list only finitely many states, never the whole infinite closure. Hence the closure is not enumerable in advance from the specification by finite means. The graph and gate are finite data (Theorem 10.1); the closure they generate is infinite; the gap is exactly the unforeseeable part. □

Remark 10.7 (Designable as instrument, not as trajectory). Theorem 10.6 says authorship reaches the standing structure and the selector but not the lived sequence: one designs the instrument, not the music. This is not a limitation to be engineered away—it is constitutive, by Theorem 8.2: the very monotonicity that individuates each state forbids pre-listing them. An authored structure is therefore necessarily open-ended relative to its author.

10.3 Bound II: an authored copy is a distinct structure

Theorem 10.8 (Non-duplication). *Let A be an existing structure with state history of committed count $m > 0$, and let A' be an authored structure specified to be isomorphic to A's standing graph and started fresh at count* 0. *Then A' is not the same structure as A: their states differ in the* *committed count, and A' satisfies a predicate ("was authored at count 0," equivalently "has empty prior history") that A does not. Resemblance of standing graph does not make the authored structure identical to the one it resembles.*

Proof. By Theorem 8.1, a state includes the committed count. A is at count $m > 0$; the fresh instantiation A' is at count 0. Hence their current states differ in the second component, $s(A) \neq s(A')$, even though their standing graphs are isomorphic. Moreover, individuation is by negation (Theorem 4.5): the two are distinguished by the partition of structures into "has prior committed history" (containing A) and "authored with empty history" (containing A'); A' is individuated by a negation—its lack of A's history—that A does not satisfy. By Theorem 8.2 no operation gives A' the prior count without taking the steps, which would make it a different walk again. Therefore $A' \neq A$: the copy is a new structure resembling the old, not the old one. $\square$

Corollary 10.9 (Authored resemblance is a new individual)**.** *An authored structure built to resemble any target is, necessarily, a distinct structure marked by its authored origin. There is no authoring that produces* the same *structure as a pre-existing one; authoring produces only new structures, some resembling old ones.*

Proof. Immediate from Theorem 10.8: the authored structure differs in committed count and in the negation that individuates it. $\square$

10.4 Bound III: authorship reaches only to the floor

Theorem 10.10 (Floor-limited authorship)**.** *An author who specifies and recovers structures using a contact graph of floor $\beta_{\mathrm{auth}} > 0$ can resolve, in any authored structure, only distinctions of boundary cost $\geq \beta_{\mathrm{auth}}$. Any sub-floor distinction—a separation of cost below β_{auth}—is not specifiable by that author and is left to the structure's own finer determinations. The author authors the parts down to the floor; the structure's sub-floor interior is its own.*

Proof. The author's specifications are themselves parts of a contact graph of floor β_{auth} (the author is a finite system; Premise 1 applies to it). By Theorem 3.1 applied to the author's graph, the author cannot represent a separation of boundary cost below β_{auth}: such a cut does not exist among the author's realisable parts. Hence any distinction the author writes into a specification has cost $\geq \beta_{\mathrm{auth}}$, and any finer distinction in the instantiated structure is below the author's resolution—present in the structure, absent from the specification. The authored structure thus has interior detail its author did not and could not specify. $\square$

Remark 10.11 (Doors, not rooms)*.* Theorem 10.10 says the author lays down the parts and their boundaries to the author's own floor—the doors—while whatever lies below that floor, the interior the structure individuates for itself, is beyond the author's reach. Together with open-endedness (Theorem 10.6) and non-duplication (Theorem 10.8), this gives the precise sense in which an authored structure is genuinely its own: foreseeable only as an instrument, never a copy, and detailed below its author's resolution.

Remark 10.12 (Interpretation: a new category, on which nothing depends)*.* The three bounds together delineate authored structures as a class with properties no specification fully governs: open-ended in trajectory, distinct under copying, and finer than their author's floor. A reader may take this as the existence of a manufactured individual that is neither foreseeable nor duplicable nor exhaustively specifiable—built, yet its own. We assert only the graph theorems (Theorems 10.3, 10.6, 10.8 and 10.10); the categorical reading is orientation and is not relied upon.

11 The Master Theorem, scope, and conclusion

We now show that the entire ladder T0–T8 is forced by, and only by, the Finiteness Premise: each rung holds for finite contact graphs and fails to be forced when finiteness or the positive floor is dropped. This is the sense in which the structures of the paper are necessary substructures of a finite world—and the sense in which structure itself is the consequence of finitude.

11.1 Where finiteness is used

Proposition 11.1 (Dependence on the premise)**.** *Each rung uses the Finiteness Premise (Premise 1) as follows:*

- (T0) *the positive floor $\beta > 0$ on edge weights, with connectedness, forces $w(\partial(U)) \geq \beta$ (Theorem 3.1);*
- (T1) *finiteness of V and the absence of a distinguished vertex force determination by complementation (Theorem 4.5);*
- (T2) *the floor forces $\rho \geq \beta$, and finiteness attains the minimising partition, giving a positive region-valued residual (Theorem 5.3);*
- (T3) *finiteness makes the isomorphism class and its partition lattice well-defined, so ρ is an attained invariant (Theorem 5.5);*
- (T4) *the floor and finiteness localise to subgraphs, giving a positive, attained private invariant (Theorem 6.2);*
- (T5) *finite degree with branching forces a gate to break finitely many ties (Theorem 7.3);*
- (T6) *discreteness of steps makes the committed count a strictly monotone natural number (Theorem 8.2);*
- (T7) *the quotient of a finite graph is a finite contact graph, inheriting the floor, so a single gate is forced on a coherent collective (Theorem 9.3);*
- (T8) *finiteness of the specification against the infinitude of the forward closure forces open-endedness, and the monotone count forces non-duplication (Theorems 10.6 and 10.8).*

Proof. Each clause restates the cited theorem's use of Premise 1; the proofs are those given in Sections 3 to 10. ☐

11.2 Failure without finiteness

We show the ladder is not forced once the premise is dropped, by exhibiting, for each load-bearing structure, a non-finite or zero-floor graph in which it fails.

Proposition 11.2 (Necessity of the premise)**.** *Drop either clause of Premise 1:*

- (a) ***Zero floor.*** *If edge weights may approach 0 ($\inf_e w(e) = 0$), then a part may be separated from the rest by a cut of arbitrarily small weight; $w(\partial(U))$ is no longer bounded below, so T0 fails, and with it the residual positivity of T2 and the gate-tie guarantees that rely on a floor.*
- (b) ***Infinite vertex set.*** *If $|V| = \infty$, a vertex may have infinite degree, so a gate must break infinitely many ties at once and the finite tie-breaking of T5 is not available in the same form; minimising partitions need not be attained, so the residual of T2 need not be realised; and a walk's "committed count" may fail to be a well-ordered monotone natural number if steps are not discrete, weakening T6.*

(c) ***Complete graph with vanishing weights.*** *On an infinite complete graph with weights tending to* 0*, every part is adjacent to every other at negligible cost: there is no positive residual, no privileged complement structure, and no forced gate—none of T0–T8 is compelled.*

Hence the substructures are consequences of the premise, not of graphs in general.

Proof. (a) Take a graph with a sequence of cuts of weight $2^{-n} \to 0$ separating a fixed part; then $\inf_U w(\partial(U)) = 0$, contradicting the conclusion of Theorem 3.1, which used $w(e) \geq \beta > 0$. (b) On an infinite star, the centre has infinite degree, so no single sub-normalised distribution over its incident edges selects after finitely many comparisons; and the partition lattice of an infinite vertex set need not attain its infimum, so $\rho(\Gamma)$ as a minimum may not exist. (c) On K_∞ with $w(e_n) = 2^{-n}$, for any part U one may route its separation through ever-cheaper edges so that $\inf w(\partial(U)) = 0$; the floor, the positive residual, and the negation-based individuation (which needs a determinate complement structure) all dissolve. Each construction violates the conclusion of the corresponding rung, so finiteness with a positive floor is necessary. ☐

11.3 The Master Theorem

Theorem 11.3 (Structure is the shadow of finitude)**.** *Let* Γ *be a graph. The substructures T0–T8 are all forced if* Γ *is a finite contact graph* (*Premise 1*)*, and none is forced if* Γ *is infinite or admits edge weights with infimum* 0*. Consequently a graph is necessarily structured—carries the floor, the negation-determined parts, the conserved region-valued residual, the private invariants, the gated trajectories, the non-returning histories, the single-gated collectives, and the bounded authorability—*if and only if *it is finite with a positive floor. Structure is neither imposed on a world from outside nor present in it independently of its parts; it is exactly the form a world of finitely many positively separated parts is compelled to take.*

Proof. The forward direction is Theorem 11.1 together with the theorems it cites: under Premise 1 each of T0–T8 holds. The reverse direction is Theorem 11.2: dropping finiteness or the positive floor defeats each load-bearing rung by explicit construction. Hence the conjunction of T0–T8 is equivalent, over graphs, to the Finiteness Premise: finite-with-floor on one side, the full ladder of structure on the other. The closing statement is the biconditional read in words. ☐

11.4 Scope

The theorems are statements about finite weighted graphs and an embedded gating datum; nothing more is assumed. The single premise (Premise 1) is a finite graph whose edges cost something. We have made no claim about any particular system. Where each rung uses the premise is marked (Theorem 11.1); that the premise is necessary is shown by counterexample (Theorem 11.2). Interpretive readings—of parts as agents, gates as selecting fields, quotient gates as collective purpose, residuals as conserved truth, private invariants as identities, authored structures as manufactured individuals—are confined to remarks (Theorems 4.8, 5.7, 6.4, 7.6, 8.5, 9.6 and 10.12) on which no theorem depends. A reader will recognise that finitely many bounded parts—of many kinds—satisfy the premise; we prove everything of the abstract graph alone, and material that could not be proved at this standard has been omitted.

11.5 Conclusion

From the single hypothesis that a world is a finite graph of positively separated parts, an entire architecture follows of necessity: no separation is free; parts are fixed by what they are not; the boundary content between parts is a conserved region that no relabelling removes and no point realises; each part carries a private conserved invariant beneath its changing trajectories; those trajectories require a selector and never return to a prior state; a coherent collective of parts has exactly one selector on its quotient; and such structures, being completely specifiable, can be authored—though authorship reaches only the instrument and not its music, produces only new individuals and never copies, and resolves only to its own floor. Each is forced by finiteness and dissolves without it. The answer to why a world is structured at all is therefore not that structure is added to it or discovered in it, but that structure is what a world *must* be for finitely many bounded parts to stand apart within it at all. Structure is the shadow of finitude; the contact graph is its form.

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